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Active Transparent Control and Assist-as-Needed Support in a Brick Breaker Exergame

A Collaborative Robotic Exoskeleton for
Rehabilitation-Style Evaluation

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Abstract and graphical abstract

This updated report describes the second-chance development of the EduExo Pro and Unity Brick Breaker rehabilitation-style platform. The previous group work created the basic game-exoskeleton interaction, in which the user controlled a Unity paddle through elbow motion and assistance was provided when the paddle was far from the predicted target. The updated work focuses on the new project objectives: transparent mode by control, real-time visual feedback, residual interaction-force evaluation, trajectory comparison over repeated sessions, and performance-based assistance recommendation.

The main correction concerns the transparent mode. In the previous implementation, the no-assistance region was treated as motor detachment or motor-off behavior. This is not equivalent to transparent control, because it removes active controllability and does not allow a smooth transition back to assistance. In the updated implementation, the servo electronics remain active and the controller applies an admittance-inspired high-compliance behavior when the user is close enough to the target. Therefore, transparent mode is implemented by control rather than by motor detachment during normal operation.

The Unity game was extended with real-time color feedback based on tracking error. Green indicates that the user is inside the transparent region, orange indicates moderate tracking error, and red indicates large tracking error. Optional low and high physical assistance levels are activated outside the transparent band. A four-level assistance intensity setting was also added: Off, Low, Medium, and High. This changes when assistance is activated, allowing lower assistance to preserve more independent movement and higher assistance to intervene earlier.

The task was reframed as a 60-second rehabilitation-style session rather than only a standard game-over condition. At the end of each session, the system displays a performance summary including tracking accuracy, mean and maximum tracking error, time in the transparent region, and use of gentle and strong assistance. Automatic logging was added to save the slider trajectory, target/ball reference, feedback state, assist level, residual torque proxy, and session summary metrics. The system was evaluated with two healthy users across four assistance intensity settings.

1. Motivation of the project

Stroke and other neuromotor impairments can reduce upper-limb control, coordination, and functional autonomy. Robotic and technology-assisted rehabilitation can support repetitive, task-oriented practice while keeping the task engaging for the user. Exergames are particularly useful because they transform repetitive therapeutic movements into a goal-directed activity with immediate visual feedback.

The Brick Breaker task was selected because it can be controlled using one degree of freedom: vertical paddle movement. This makes it compatible with the EduExo Pro elbow flexion-extension motion. The game also provides a clear target reference: the user must move the paddle toward the expected ball interception point. This enables an assist-as-needed strategy in which the user remains the primary motion generator, while the exoskeleton intervenes only when performance indicates that support may be useful.

In the updated version, the game is therefore treated not only as a win/loss task, but as a short timed rehabilitation session in which movement quality, tracking performance, assistance use, and transparent interaction can be measured.

2. Objective of the work

The objective of the updated work is to transform the previous game-exoskeleton prototype into a more complete rehabilitation-style evaluation platform. The update focuses on the requirements communicated for the second-chance examination: transparent mode must be implemented by control rather than by motor detachment; the game must provide feedback when the user is far from the catching position; and the report must include residual torque evaluation and comparison plots between the slider trajectory, target/ball position, and feedback state over repeated games.

- Maintain the previous EduExo-Unity interaction, where elbow angle controls the Brick Breaker paddle.
- Implement **active transparent** control in the safe region without using motor detachment as the normal transparent behavior.
- Add **visual color feedback** based on tracking error:
 - green = safe / inside transparent region,
 - orange = moderate tracking error,
 - red = large tracking error.
- Keep low and high physical assistance as an **optional assist-as-needed** extension outside the transparent region.
- Log residual torque proxy, trajectory, feedback, and assist states to allow quantitative evaluation.
- Evaluate performance across multiple assistance intensity settings and use the results to guide future assistance reduction.



Fig. 1 Unity Brick Breaker game interface used for the EduExo rehabilitation task.

3. Requirements

3.1 Functional requirements.

TABLE 1

Requirement	Description of the requirement	Verification proposal
Control of the game	The user must control the Brick Breaker paddle using voluntary elbow movement measured by the EduExo Pro angle sensor.	Compare the measured arm angle with the Unity slider/paddle trajectory during a complete session.
Transparent mode by control	When the paddle is inside the transparent region, the motor must remain active and controlled, without normal motor detachment.	Verify that <code>assistLevel = 0</code> corresponds to a neutral/high-compliance command and log residual interaction force during this state.
Assist-as-needed control	The system must provide low or high support when the user is far from the target and assistance is enabled.	Compare assistance activation against tracking error and target distance.
Real-time visual feedback	The game must communicate the distance from the target using color feedback.	Observe and log feedback state: Safe, Medium, and Far.
Assistance intensity adjustment	The user must be able to select the global assistance intensity: Off, Low, Medium, or High.	Verify that each setting changes the transparent/assist thresholds and is reflected in the logged assist levels.

3.2 Non-functional requirements.

TABLE 2

Requirement	Description of the requirement	Verification proposal
Safety	The exoskeleton must operate within comfortable movement and command limits.	Use limited servo command ranges, safe neutral/high-compliance behavior, and emergency power buttons.
Physiological range of motion	The mapped game range must correspond to the practical strapped arm range of the user.	Calibrate the angle sensor using measured raw values for the comfortable strapped range.
Usability	The setup should be sufficiently simple for repeated short rehabilitation sessions.	Use a 60-second timed session mode, pause menu, assistance slider, and end-session performance summary.
Reliability	The system should run without communication failures during sessions.	Use USB Serial logging/communication during evaluation to avoid UDP instability during testing.
Transparency	The controller should minimize interaction force during transparent operation.	Evaluate mean and maximum residual torque proxy during transparent mode.

4. Materials and methods

4.1 Hardware

The hardware platform is the EduExo Pro, configured as a one-degree-of-freedom elbow exoskeleton for flexion and extension. The user wears the device on the arm and controls the game paddle by moving the elbow. The Arduino reads the angle sensor and the wrist interaction force sensor. The angle signal is mapped to a normalized 0-100 range and then converted by Unity into the vertical paddle position.

A force sensor at the wrist is used to estimate the residual interaction force between the user and the exoskeleton. Because the device does not contain a direct joint torque sensor, the report uses the force sensor as a residual torque proxy. This is expressed as a percentage of the full sensor scale for interpretability.

4.1.1 Communication

The original group implementation used WiFi UDP communication. During the updated individual work, a wired USB Serial connection was adopted for the evaluation sessions because it was more reliable and easier to debug within the limited laboratory time. Arduino sends the current angle, target angle, assist level, residual torque proxy, servo command, raw angle, and raw force values. Unity sends back the assist level and target angle as a command.

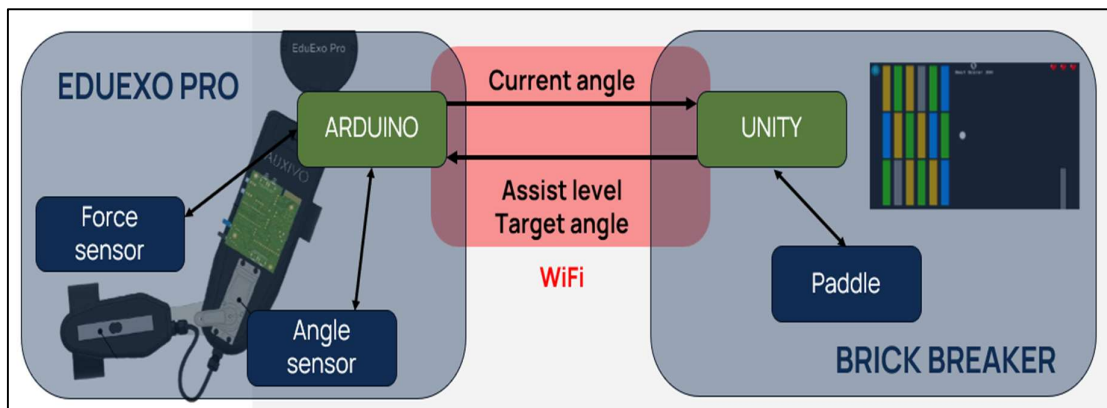


Fig. 2 Original WiFi-based EduExo Pro–Unity communication architecture from the group implementation.

TABLE 3- SERIAL COMMUNICATION VARIABLES EXCHANGED BETWEEN ARDUINO AND UNITY

Direction	Data sent	Purpose
Arduino to Unity	currentAngle, targetAngle, assistLevel, residualTorquePercent, servoCommand, rawAngle, forceRaw	Move the paddle, display debug values, and log evaluation data.
Unity to Arduino	assistLevel, targetAngle	Select transparent, low-assist, or high-assist behavior and define the target reference.

4.2 Kinematic and dynamic model

The EduExo Pro is modelled as a one-degree-of-freedom elbow exoskeleton supporting elbow flexion and extension. The shoulder is not actively controlled in this work, and the relevant controlled variable is the elbow joint angle, θ . The user's elbow angle is measured by the EduExo angle sensor and mapped to a normalized range from 0 to 100. Unity then converts this normalized value into the vertical position of the Brick Breaker paddle.

From a kinematic point of view, the arm can be approximated as a planar single-link system, where the wrist attachment point depends on the elbow angle and on the forearm segment length, L . A simplified forward kinematic representation can be written as:

$$\rho(\theta) = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} L \cos \theta \\ L \sin \theta \end{bmatrix}. (1)$$

where ρ is the position of the wrist attachment point. The corresponding end-effector velocity is related to the elbow angular velocity through the Jacobian:

$$\dot{\rho} = J(\theta)\dot{\theta}, \quad J(\theta) = \begin{bmatrix} -L \sin \theta \\ L \cos \theta \end{bmatrix}. (2)$$

In the implemented game, this full Cartesian model is not directly used for control. Instead, the elbow angle is used as the main input variable and is mapped to the paddle position. The target reference in Unity is defined by the predicted ball-catching position or target trajectory. The tracking error is computed as the difference between the user-controlled paddle position and this target reference. This error is then used to define the feedback state and the assistance level.

The dynamic behavior of the elbow-exoskeleton system can be described conceptually as a single-joint rotational system, where the required joint torque depends on inertia, velocity-dependent terms, and gravity:

$$\tau = I \cdot \alpha + C(\theta, \dot{\theta}) \cdot \dot{\theta} + G(\theta). \quad (3)$$

where τ is the joint torque, I is the rotational inertia, α is the angular acceleration, C represents velocity-dependent effects, and G represents gravity. However, the implemented system does not perform full model-based torque control, because the EduExo Pro does not provide a direct joint torque sensor. Therefore, this equation is used only as a conceptual reference for the dynamics.

For the updated project, transparency is evaluated experimentally using the wrist force sensor. The force sensor measures the interaction force between the user and the exoskeleton. Since this is not a direct elbow torque measurement, the measured force deviation from its calibrated rest offset is used as a residual torque proxy. This proxy is expressed as a percentage of the selected force-sensor scale. Lower values indicate that the transparent controller allows the user to move with less unwanted resistance, while peaks indicate moments of higher interaction force, friction, or imperfect tuning.

Because the EduExo Pro is stiffly actuated, transparent behavior is mainly produced through control. In transparent mode, the actuator remains active and the controller applies a neutral or high-compliance admittance-inspired behavior instead of detaching the motor. Outside the transparent region, optional low or high assistance can be activated depending on the tracking error and the selected assistance intensity.

4.3 Control Design

4.3.1 Hardware level of compliance

The EduExo Pro is a stiffly actuated exoskeleton and does not include intrinsic mechanical compliance such as a series elastic actuator. For this reason, transparent interaction must be created mainly through software control rather than through passive mechanical elasticity. In the updated system, compliance is produced by limiting the command range, keeping the motor active in transparent mode, and using a neutral/high-compliance behavior when the user is inside the transparent region.

The emergency power buttons are kept as a safety mechanism, but they are not used as the normal implementation of transparent mode. This distinction is important: transparent control should allow the user to move freely while the actuator remains controllable, whereas motor detachment removes active control and makes transitions back to assistance less smooth.

4.3.2 Low-level control solution

The low-level controller is admittance-inspired. The force sensor measures the interaction between the user and the device. After subtracting the calibrated force offset, the remaining force signal is used either to support high-compliance transparent behavior or to provide low/high assistance when the user is outside the transparent band.

In transparent mode, the motor electronics remain active and the controller sends a neutral or high-compliance command. The goal is to minimize resistance while preserving controllability. The force-sensor deviation from its calibrated offset is also logged as a residual torque proxy:

$$\text{residualTorqueProxy} = |\text{forceRaw} - \text{forceOffset}|$$

For easier interpretation, this value is expressed as a percentage of the selected full-scale force-sensor range. Since the EduExo Pro does not include a direct elbow torque sensor, this value should be interpreted as a residual interaction-force proxy rather than an exact joint torque measurement.

When assistance is enabled, the controller uses the assist level sent by Unity. The low-assist condition provides gentle support, while the high-assist condition provides stronger support. The assistance command is limited around the neutral motor command to avoid excessive stiffness or unsafe motion.

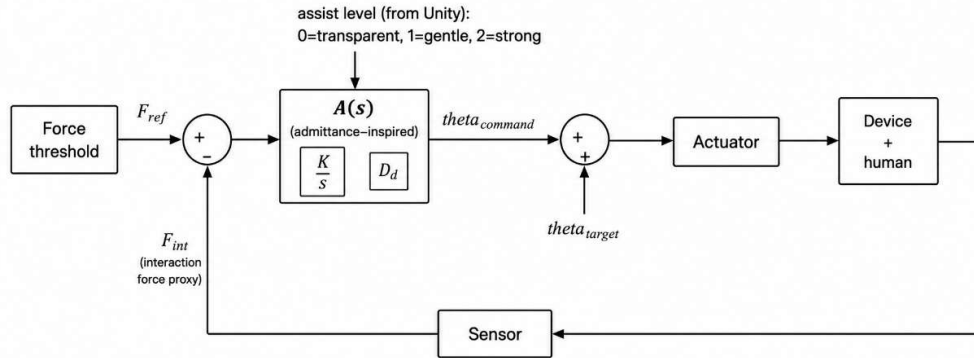


Fig. 3 Low-level admittance-inspired control architecture.

The diagram represents the admittance-inspired low-level control structure used in the updated implementation. The wrist force sensor provides an interaction-force signal, which is compared with a calibrated offset and filtered by a threshold. Depending on the assist level sent by Unity, the controller either applies active transparent behavior or low/high assistance. Since no direct joint torque sensor is available, the measured force is interpreted as a residual interaction-force proxy rather than an exact elbow torque.

4.3.3 High-level control solution

The high-level controller is implemented in Unity. Unity computes a target reference from the predicted ball-catching position and compares it with the actual user-controlled paddle position. A transparent band is defined around the target trajectory. If the paddle remains inside this band, Unity sends `assistLevel = 0` and the Arduino applies transparent control. If the paddle moves outside the transparent region, the ball color changes to communicate the tracking error, and optional assistance can be activated.

TABLE 4 - THE THREE STATES OF FEEDBACK AND ASSISTANCE LOGIC

State	Condition	Unity feedback	Arduino behavior
Transparent / Safe	Paddle inside transparent band	Green	<code>assistLevel = 0</code> ; motor active; neutral/high-compliance command
Medium error	Paddle moderately outside band	Orange	optional low assistance if enabled
Large error	Paddle far from target	Red	optional high assistance if enabled

In active transparent mode, the user remains the only source of voluntary motion and the robot minimizes resistance. In active-assistive mode, the user still contributes voluntary effort, but the robot adds low or high support depending on the tracking error.

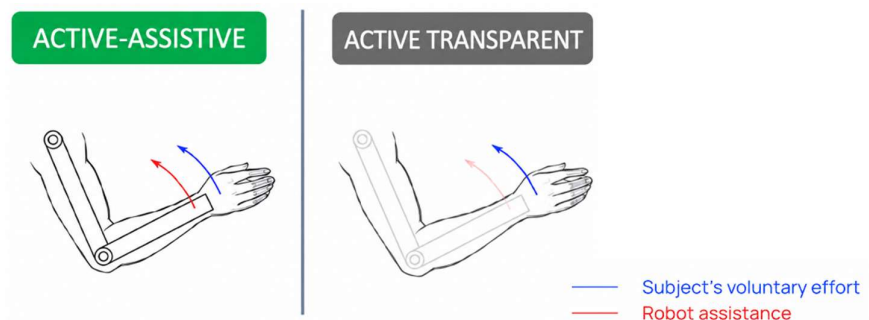


Fig. 4: High-level control modes: active transparent mode and active-assistive mode.

4.3.4 Error-based assist-as-needed strategy

A four-level assistance intensity setting was added in Unity: Off, Low, Medium, and High. This setting changes the thresholds used to decide when the system remains in transparent mode and when assistance is allowed.

In the Off condition, physical assistance is disabled and the system remains in transparent mode while still providing visual feedback. In the Low condition, the transparent band is wider, so assistance is activated only when the tracking error becomes large. In the Medium and High conditions, the transparent band becomes progressively smaller, meaning that assistance activates earlier.

This structure preserves the assist-as-needed principle. The user remains the primary motion generator, while the exoskeleton provides additional support only when the tracking error indicates that assistance may be useful.

The user's intention is not estimated from muscle activity in this implementation. Instead, it is inferred indirectly from measurable signals available in the system: elbow angle, paddle position, tracking error, and interaction force from the wrist force sensor. The force sensor is used to detect interaction with the exoskeleton and to support the admittance-inspired controller, while the tracking error determines the feedback state and optional assistance level.

4.4 Protocol for testing and validation

4.4.1 Component Test Protocol

The angle sensor was calibrated by measuring the practical strapped-arm range of the user. The raw angle sensor signal was mapped to a normalized 0–100 range, corresponding to the comfortable usable movement range during the experiment. The force sensor was calibrated at rest to define a force offset, so that subsequent readings represented the interaction force between the user and the exoskeleton rather than the sensor baseline. The servo command was also tested around the neutral value to identify the command that minimized perceived resistance during transparent mode.

4.4.2 System Test protocol

The integrated system was tested by launching Unity, opening the USB Serial connection, and verifying that the paddle moved according to the EduExo angle measurement. The communication loop was checked by confirming that Arduino sent the current angle, assist level, residual torque proxy, servo command, raw angle, and raw force values, while Unity sent the target angle and selected assist level back to Arduino.

Unity was also checked to ensure that it logged the target position, actual slider/paddle position, feedback state, assist level, residual torque proxy, and session summary metrics. This

confirmed that the required data pipeline was available for trajectory comparison, feedback-state analysis, assistance-state analysis, and residual torque evaluation.

4.4.3 Experimental User Test Protocol

Two healthy users performed 60-second rehabilitation-style sessions using the EduExo-controlled Brick Breaker game. Four assistance intensity settings were tested: Off, Low, Medium, and High. To reduce the effect of learning or fatigue, the two users tested the assistance settings in opposite orders: one user started from Off and progressed toward High, while the other user started from High and progressed toward Off.

Each session produced a CSV file and a summary table. The main outcome variables were tracking error, time spent in the transparent region, residual torque proxy during transparent mode, and the number of assistance activations. Completed sessions were used for the main quantitative comparison, while failed or intentionally interrupted sessions were treated as additional observations rather than primary evaluation trials.

5. Results

5.1 Study population and Task Overview

The updated evaluation was performed with two healthy users. The task was reframed as a timed rehabilitation-style session rather than only a standard game-over condition. Each session lasted 60 seconds, during which the user controlled the Unity paddle with the EduExo Pro and attempted to remain close to the target/ball reference while receiving real-time visual feedback and optional physical assistance.

Four assistance intensity settings were tested: Off, Low, Medium, and High. To reduce the influence of learning or fatigue, the two users tested the settings in opposite orders. User 1 performed the sessions from Off to High, while User 2 performed the sessions from High to Off. Each session generated a CSV file and a summary file containing trajectory, feedback, assistance, residual torque proxy, and performance metrics.

TABLE 5 - STUDY POPULATION AND EXPERIMENTAL SESSION OVERVIEW.

Parameter	Value
Participants	2 healthy users
Session duration	60 seconds
Assistance settings	Off, Low, Medium, High
Logged frequency	Approximately 20 Hz
Main outputs	Trajectory, feedback state, assist level, residual torque proxy, summary metrics

5.2 Functional requirements: Game completion and Real-Time Visual Feedback

The Unity interface provided real-time color feedback based on the distance between the user-controlled paddle and the target reference. Green indicated that the paddle was inside the transparent region, orange indicated moderate tracking error, and red indicated large tracking error. This feedback remained available independently of physical assistance, so the user could still self-correct even when assistance was disabled.

Figure 5 shows one representative session. The top plot compares the paddle trajectory with the target/catching position and the transparent band. The bottom plot shows the logged feedback state and assist level over time. This verifies that the visual feedback changed according to tracking error and that the corresponding assistance states were correctly recorded.

Trajectory comparison and Shown Feedback

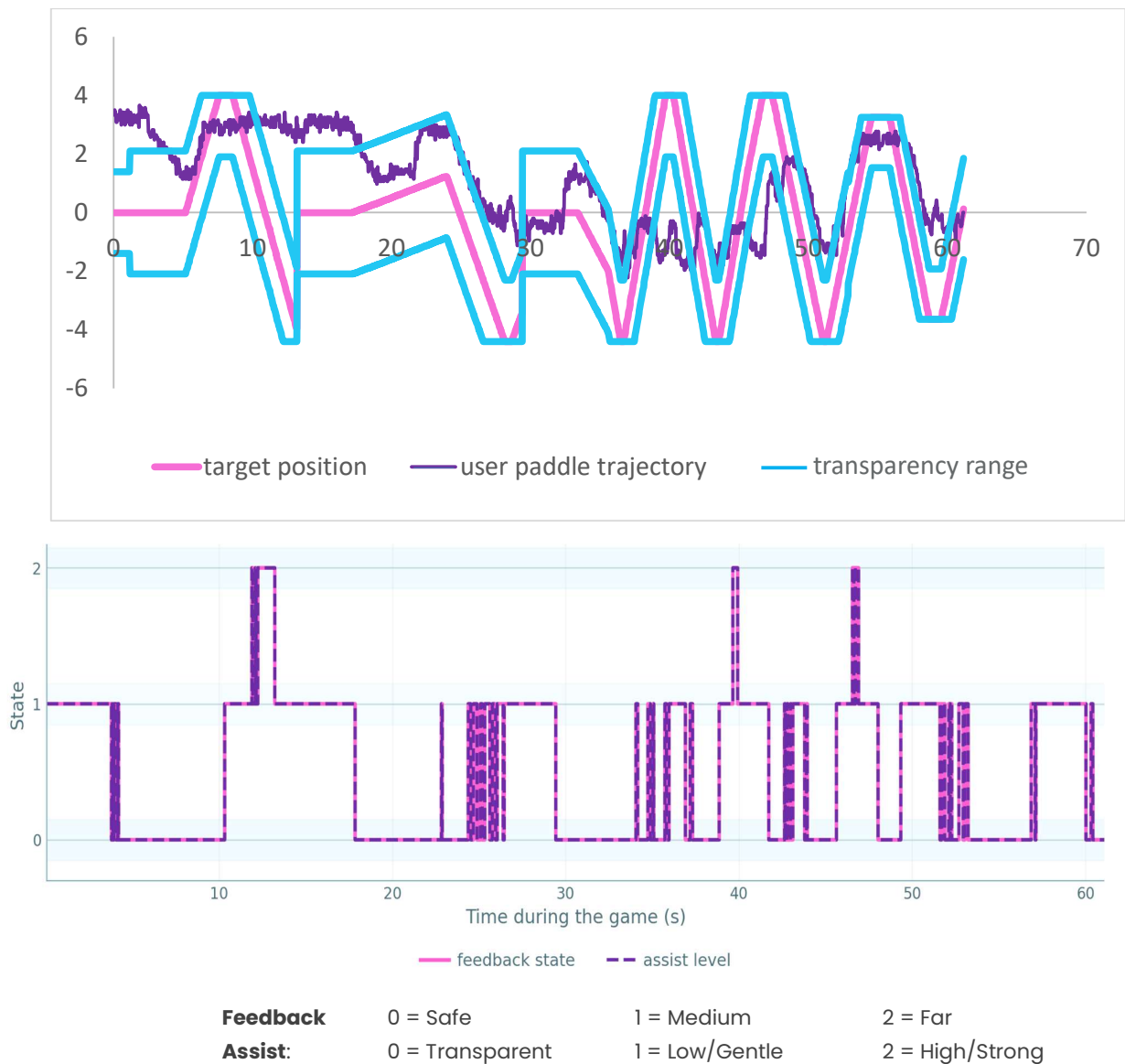


Fig. 5: Representative session showing target position, user paddle trajectory, transparent band, feedback state, and assist level over time.

5.3 Functional requirements

5.3.1 Control of the game and assist-as-needed control

The system preserved the assist-as-needed logic by changing assistance behavior according to the selected assistance intensity setting. In the Off condition, physical assistance was disabled and the system remained in transparent mode while still providing visual feedback. In the Low, Medium, and High conditions, assistance was activated progressively earlier as the user moved farther from the target reference.

Figure 6 shows the percentage of time spent in each assistance state. As expected, the Off condition remained fully transparent. Low assistance preserved mostly independent movement, while Medium and High assistance increased the time spent in gentle or strong assistance. This confirms that the assistance intensity slider successfully modified the control behavior.

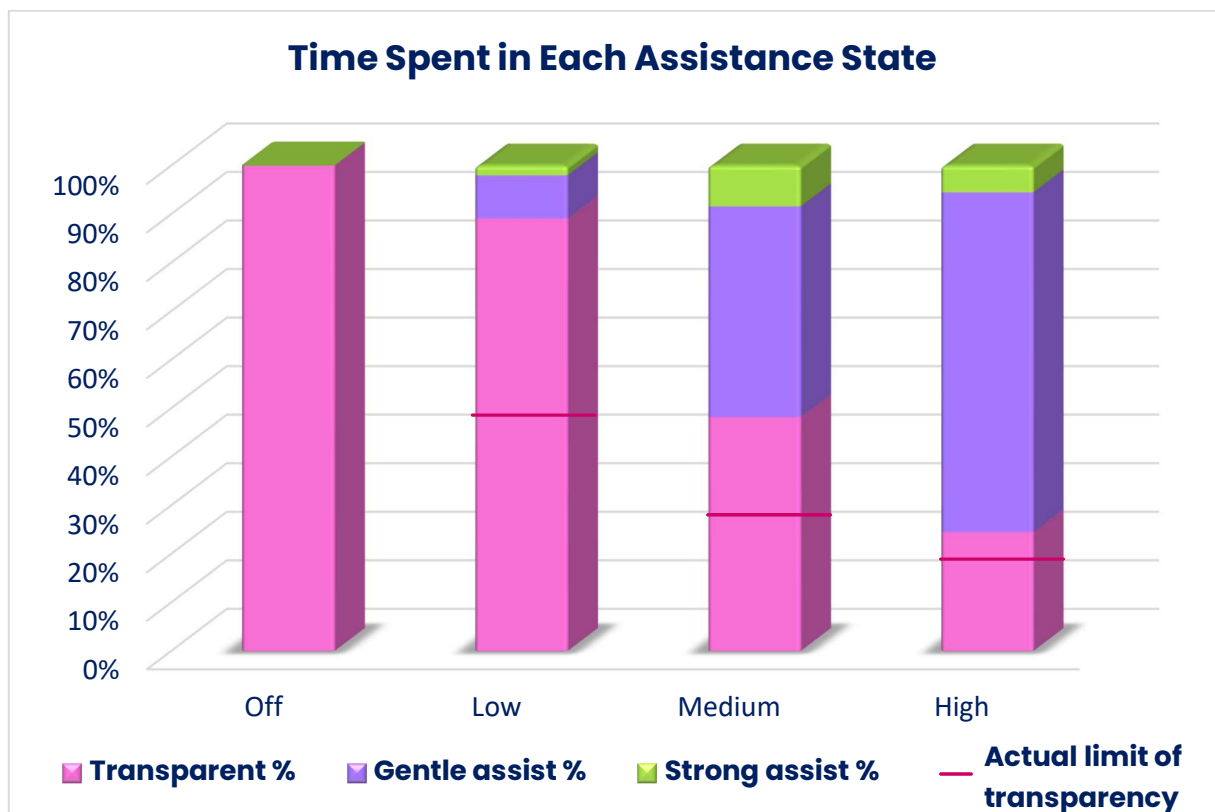


Fig. 6: Percentage of time spent in transparent, gentle-assist, and strong-assist states across assistance settings.

5.3.2 Task accuracy and performance metrics

Task accuracy was quantified using the tracking error between the target reference and the actual paddle trajectory. The results show that assistance intensity changed the system behavior, but did not produce a perfectly linear improvement in accuracy. This is expected in a human-in-the-loop task, because performance depends not only on the assistance level but also on the user's strategy, reaction time, and adaptation to the exoskeleton.

Figure 7 summarizes tracking accuracy, mean tracking error, and independent movement time across the four assistance settings. The clearest trend is observed in independent movement time: as assistance intensity increased, the percentage of time spent in the transparent region decreased. This shows the main rehabilitation trade-off: higher assistance provides more guidance, while lower assistance encourages more autonomous movement.



Fig. 7: Performance metrics across assistance intensity, including tracking accuracy, mean tracking error, and independent movement time.

The summary metrics in Table 6 further confirm this trend. Transparent time decreased from 100.0% in the Off condition to 29.2% in the High condition, while assistance activations

increased from 0 to 64. This confirms that the assistance settings changed when and how often the robot intervened.

TABLE 6 - SUMMARY METRICS ACROSS ASSISTANCE SETTINGS

Game ID	Assistance setting	% time in transparent region	Mean tracking error %	Max tracking error %	Mean residual torque %	Max residual torque %	Assistance activations
Game 1	<i>Off</i>	100.0	28.8	93.0	4.2	46.2	0
Game 2	<i>Low</i>	91.8	18.9	72.0	3.7	16.6	9
Game 3	<i>Medium</i>	51.7	25.5	85.9	3.4	18.2	48
Game 4	<i>High</i>	29.2	19.7	58.0	3.8	20.8	64

5.4 Interaction forces and residual torque proxy

The wrist force sensor was used to estimate the residual interaction force between the user and the exoskeleton. Since the EduExo Pro does not include a direct elbow joint torque sensor, the measured force-sensor deviation from its calibrated rest offset was used as a residual torque proxy. Therefore, the reported values should not be interpreted as exact biomechanical joint torque, but as an indicator of the remaining resistance felt by the user during interaction.

For the evaluation of transparent control, the residual torque proxy was analysed mainly during transparent mode, where assistLevel = 0. This allows the transparency of the controller to be assessed independently from the low/high assistance states. A low mean residual torque proxy indicates that the user could move with limited resistance, while high peaks reveal short moments of increased interaction force, friction, imperfect alignment, or non-ideal tuning.

Figure 8 shows the mean and maximum residual torque proxy during transparent mode for three representative games.

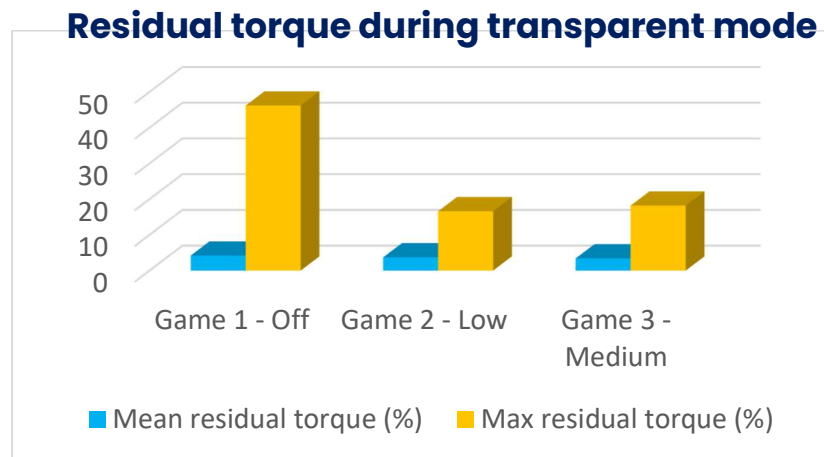


Fig. 8: Mean and maximum residual torque proxy during transparent mode across three representative games.

5.5 Requirements: Transparency of the robot and safety

The updated transparent mode addresses the central requirement of the second-chance project: the motor is not detached during the normal transparent state. Instead, the servo electronics remain active and the control command is kept neutral or high-compliance. The residual torque proxy is then logged to evaluate whether the user still experiences significant resistance during transparent interaction. Low mean residual torque values indicate that the user can move with limited interaction force, while peaks reveal moments where tuning, friction, alignment, or user-device interaction still require improvement.

Safety remains supported by physical power buttons, limited servo command ranges, and calibrated movement limits. Importantly, the emergency power-down function is separate from transparent mode. Motor detachment or power-down is kept only as a safety mechanism, not as the normal implementation of transparency.

5.6 Non-Functional requirements

The updated system improves usability by providing a 60-second timed session mode, an end-session performance summary, and automatic CSV logging. The game-over behavior was modified so that even if the user loses before the timer ends, the rehabilitation performance summary is still shown. This is important because losing the game does not make the rehabilitation data invalid.

The pause and end-session menus allow the assistance intensity to be changed between Off, Low, Medium, and High, making it easier to repeat sessions at different difficulty levels. Reliability was also improved during evaluation by using USB Serial communication instead of WiFi UDP, reducing communication uncertainty and allowing stable logging of the required variables. Overall, these changes make the system more suitable for repeated short rehabilitation-style sessions and for quantitative post-processing.

6. Discussion and conclusion

6.1 Final Discussion on Results

The updated work directly addresses the main limitation identified in the previous implementation. The original assist-as-needed concept was preserved, but the close-to-target region was redesigned as an active transparent mode rather than a motor-detached state. This makes the control strategy more coherent with transparent interaction: the system remains controllable, transitions smoothly back to assistance, and allows residual interaction force to be evaluated.

The results show the expected rehabilitation trade-off. Lower assistance settings preserved a larger percentage of transparent movement and encouraged user autonomy. Higher assistance settings increased intervention, which can help guide the user when tracking error becomes large, but also reduced independent movement time. This supports the idea that assistance should not simply be maximized, but adapted according to the user's performance and rehabilitation goal

6.2 Communication choice

Although the original implementation used WiFi UDP communication, the final evaluation used USB Serial communication for reliability during data acquisition. This was a pragmatic design choice for the examination work: the wired connection reduced communication uncertainty, simplified debugging, and allowed stable logging of the required evaluation variables. A future version can return to optimized wireless communication once the control and logging pipeline is fully validated.

6.3 Active Support

The low and high assistance levels act as optional support outside the transparent region. The assistance branch should not be interpreted as the definition of transparent control; instead, it is an extension of the base transparent controller. In transparent mode, the user remains the primary motion generator and the robot minimizes resistance. When tracking error becomes too large, the controller can add gentle or stronger support depending on the selected assistance intensity.

6.4 System Limitations

The main limitation is the indirect estimation of residual torque. The force sensor provides a useful residual interaction-force proxy, but it does not measure elbow joint torque directly. Therefore, the reported residual torque values should be interpreted as comparative indicators of transparency rather than exact biomechanical torque measurements.

In addition, transparent behavior is sensitive to the neutral servo command, the physical fit of the exoskeleton, the strapped arm range calibration, and the selected compliance gains. Further tuning is required to make transparent mode feel consistently light while maintaining enough assistance when needed. The evaluation was also performed with healthy users, so the results represent a technical validation rather than a clinical validation. Users with impaired motor control may require patient-specific ranges, lower gains, slower transitions between assistance states, and longer familiarization periods.

6.5 Future Development

- Further tune the neutral command and compliance gains to reduce residual interaction force during transparent mode.
- Improve force-to-torque estimation by including the lever arm and a more complete sensor calibration.
- Implement patient-specific adaptation of the transparent band and assistance thresholds.
- Use performance metrics such as mean tracking error, tracking accuracy, and time in transparent mode to automatically recommend or reduce assistance when the user improves.
- Improve the user interface of the end-session performance summary and assistance selection screen.
- Validate the platform with users with neuromotor impairments and longer rehabilitation sessions.

6.6 Conclusion

The previous EduExo–Unity Brick Breaker prototype was extended into a more complete rehabilitation-style evaluation platform. The main contribution of the updated work is the implementation of active transparent control without using motor detachment as the normal no-assistance behavior. The system now provides real-time visual feedback, optional low/high assistance, a four-level assistance intensity setting, timed rehabilitation sessions, end-session performance summaries, and automatic data logging.

The logged results allow the system to be evaluated quantitatively through trajectory comparison, feedback-state analysis, assistance-state distribution, residual torque proxy, and performance metrics. Overall, the updated platform demonstrates a more coherent assist-as-needed rehabilitation approach: the user remains the main source of movement, while the robot provides measurable and adjustable support only when needed.

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